



**Department of Science,
Engineering, and Mathematics**
333 Murfreesboro Road
Nashville, TN 37210

**Syllabus
EEC 3150-31 Engineering
Programming II**

Fall 2020
Monday / Wednesday / Friday
12:00 p.m. – 12:50 p.m.
Online Academic Week:
Monday 12:01 a.m. – Sunday 11:59 p.m.

Professor: Timothy S. Myatt

Office Information

Office: BBB 145
Office Hours: Open door or by appointment (preferred)

Contact Information

Email: tmyatt@trevecca.edu
Office phone: (615) 248-1560
Cell phone: (615) 933-9887

Course Description / Overview

Description – 3 credit hours

This course makes use of the Python programming language to design object-oriented software for engineering applications. Topics include classes, encapsulation, inheritance, composition, polymorphism, file I/O, and Graphical User Interfaces. Additional topics include numerical methods and data visualization using Python.

Prerequisite(s)

EEC 2150 Engineering Programming I.

Course Objectives and Student Learning Outcomes

The general objective of the Electrical and Computer Engineering Major of the Department of Science, Engineering, and Mathematics is to prepare graduates with a total learning experience by exposing the student to the entire spectrum of electrical and computer engineering within a caring Christian environment. To prepare students with a strong technical foundation for entering the global workforce, seeking advance education, and serving within their communities, motivated by Christian ideas. The specific objective of this intermediate course in computer programming is to advance skills in algorithmic design in an object-oriented environment. The Python programming language will be used to advance these skills.

The course will focus substantially on hands-on lab work. In addition to advancing problem-solving skills using computational methods, participants will further develop analytical and communication skills that are essential to success in today's engineering environments.

After completion of this course, it is expected that students will increase their ability to:

1. Express solutions to problems computationally using the Python programming language
2. Recognize challenges and solutions in computational and algorithm complexity
3. Map scientific problems into computational frameworks based on object-oriented design
4. Take a systematic approach to organizing, writing, and debugging non-trivial software

Course Resources

Required resources

1. Guttag, J. V. (2016). *Introduction to Computation and Programming Using Python with Application to Understanding Data (2nd ed.)*. Cambridge, MA: The MIT Press. ISBN: 9780262529624
2. A notebook computer that meets the minimum specifications for software development.

Supplemental resources

1. TNU learning management system from Blackboard: <https://trevecca.blackboard.com>.
A Blackboard course may also include one or more of the following integrations:
 - **LinkedIn Learning:** An online video library of engaging, top-quality courses taught by recognized industry experts. It is free through a Blackboard integration.
 - **Smarthinking;** Online Tutoring has limited technical tutoring, but the Online Writing Lab provides students with critiques of any written assignment. It is free through Blackboard integration (see <https://www.pearson.com/us/higher-education/products-services-institutions/smarthinking-online-tutoring.html#>)

Course Policies

This course is subject to the policies, procedures, and guidelines described in the University Catalog. It includes, but is not limited to official statements about copyright, intellectual property, academic integrity, class attendance, assessment and grades, records and registration, and academic requirements. (See <http://www.trevecca.edu/academics/academic-catalog>). Statements below supplement but do not supersede the catalog.

Scheduled Class Sessions

This course is scheduled to meet 20 times for 50 minutes in a face-to-face (F2F) format (not including the final exam). Students are expected to attend at the scheduled time.

Time Commitment

The average student can expect to invest approximately 15-16 hours per week in course activities for a semester total of 112 hours of learning activities.

Course Communication

There are two official forms of communication for this course: 1) Blackboard messages, and 2) Trevecca email. It is recommended that these forms of communication be consulted no less than four times per week. Blackboard messages is the preferred.

Attendance Policy

Attendance is an important aspect of this course. For the Fall 2020 semester, students will be allowed to attend this class either in person (classroom attendance) or remotely via Blackboard and live stream video (where applicable).

Students should NOT attend class in person, and are expected to attend class remotely, if they A) have chosen to live off-campus as a remote student, B) are experiencing any of the symptoms associated with COVID-19, C) have been in close contact with someone who has tested positive for COVID-19, or D) are experiencing ANY symptoms of illness (sore or scratchy throat, fatigue, itchy eyes, etc.).

A student will be marked absent for a class session if he or she does not attend the class in person (physically in the classroom), does not participate in the live-streamed class session (if applicable), AND does not participate in non-live (asynchronous) Blackboard classroom activities prior to the next time the class meets physically. Students should make every effort to attend, and participate in, the live-streamed class experience at the regularly scheduled class time (if provided), but if this is not possible, they **MUST** view the video from the classroom experience and/or participate in Blackboard course activities to be counted present.

Excused absences are defined only as an extended illness verified by a licensed medical professional, participation in an approved University related event, or another legitimate cause as determined by the instructor. Students will be required to make-up any work that is missed due to an excused absence, but the instructor may give the student additional

time to complete the assignment. Students should notify their instructor of absences that may be considered excused in advance when possible.

University policy states that the total number of excused and unexcused absences for a student cannot exceed 20% of the total class time. The *University Catalog* states, “If a student has missed more than the allowed number of absences in a given course for any reason, the student automatically fails the course unless the student officially withdraws from the course...” NOTE: The university’s grade appeal policy, located in the catalog, will remain available for all students this semester.

Missed Due Dates

Academic submissions past the due date will not be accepted and will result in a grade of zero. This includes all submitted work, including quizzes. No exceptions.

Electronic Devices

The use of electronic devices during the scheduled class times is restricted to those activities directly related to the course. It would be a challenge to imagine a scenario when a smartphone would be used for activities directly related to the course if not expressly requested to do so. The professor, at his discretion, may judge an activity unrelated, at which point a “cognitive absence” will result. Students should only use their electronic devices for course related activities.

Electronic Submissions

Assignments must be submitted electronically (digitally). It is the student’s responsibility to ensure that the professor receives the assignment in the correct format and on time. All digital exchanges are submitted freely by the student and the university reserves the non-exclusive right to use work submitted to further the purposes of the university.

Trevecca Nazarene University Recorded Lecture Policy

Faculty members are permitted to video and/or audio record class sessions provided that recordings are posted in a secure environment and are only available to students registered for the course. Students who have a qualifying disability are also allowed to record class sessions for their personal study only. Information contained in the recorded lecture is protected under federal copyright laws and may not be published without the consent of the instructor. As video and/or audio of students may be captured as a part of the recording, it is also a Family Educational Rights and Privacy Act (FERPA) violation to share classroom recordings. Lectures/class sessions recorded for any reason may not be shared with anyone who is not a registered participant of the course without the consent of the instructor.

Academic Honesty Statement

Academic honesty is expected of all students at Trevecca Nazarene University and students are held to high standards of ethics and integrity. It is an integral part of the educational process where learning takes place in an atmosphere of mutual trust and respect. Each student is responsible for maintaining high standards of academic ethics, personal honesty, and moral integrity. Dishonest academic behavior will be dealt with

fairly and firmly. Violations of the policy may result in the immediate failure of the course. Students can visit the Student Handbook for more specific examples of infractions and guidelines for dealing with academic dishonesty.

Faculty members are expected to report all incidents of academic dishonesty to the Associate Provost and Dean of Academic and Student Affairs. Academic dishonesty may result in a failing grade in a course, or, in cases of multiple infractions, dismissal from the University.

Collaboration / Group Assignments

Collaboration is an important part of the learning process. Discussing assignment problems and their solutions is encouraged. However, even though the process of problem solving is collaborative, copying another student's work or copying from other sources is academically dishonest. Because quizzes and exams are not intended to be collaborative, solicitation of assistance is strictly prohibited.

Disability Statement

At Trevecca Nazarene University, we strive to make learning experiences as accessible as possible. If you anticipate or experience barriers based on disability, please contact Disability Services at 615-248-1463 or email disabilityservices@trevecca.edu to establish reasonable accommodations. Disability Services is located in the Bud Robinson Building, lower level. Trevecca Nazarene University complies with Section 504 of the Rehabilitation Act of 1973 and the Americans with Disabilities Act of 1990, as amended by the ADA Amendments Act of 2008.

Class Cancellation

Students will be informed by established methods if a class must be cancelled.

Feedback

Questions and comments about any aspect of the course are welcomed and encouraged. Constructive criticism will benefit present and future classes. TNU and the professor are committed to continuously improving the content and methods used. Feedback will never affect a student's grade.

Course Activities

Readings

Each week several chapter sections in the text will be required reading with LinkedIn Learning videos as supplemental. Additional online documentation is extensive.

Quizzes / Exams

Two unannounced quizzes, one mid-term exam (in two parts), and one final exam will form the foundation for assessment supplemented by the hands-on assignments.

Hands-on Assignments (HOA)

The thirteen hands-on lab exercises will be a significant part of the learning process. They are intended to be collaborative and working in groups is encouraged. However, students must submit their own work (see Academic Honesty Statement above).

Course Assessment

Grading

Course Activity	Points	Percent
Quizzes (2 * 25)	50	5.0 %
Exams (2 * 100)	200	20.0 %
Hands-on Assignment-0	30	3.0 %
Hands-on Assignment 1-12 (12 * 60)	720	72.0 %
Total	1000	100.0 %

Grading Scale			
Excellent	A+ : 97 – 100%	A : 93 – 96.9%	A- : 90 – 92.9%
Good	B+ : 87 – 89.9%	B : 83 – 86.9%	B- : 80 – 82.9%
Average	C+ : 77 – 79.9%	C : 73 – 76.9%	C - : 70 – 72.9%
Below Average	D+ : 67 – 69.9%	D : 63 – 66.9%	D- : 60 – 62.9%
Failing	F : below 60%		

Assessment and Objectives

Objective	
1-4	Quizzes / Exams
1-4	Hands-on Assignments

Distribution of Student Learning Hours

Assignments	Total Learning Hours
Readings / Lectures / Training	42
Quizzes / Exams	10
Hands-on Assignments	60
Total	112

Course Schedule

NOTE: The course schedule exists in a separate document.

Syllabus Disclaimer & Notice of Changeability

The syllabus is intended to act as a guiding document for the course. It establishes the good faith intentions of the professor to provide the student with the learning objectives, resources, policies, activities, schedule, and method of assessment that will maximize student success. Although it is not a binding legal contract, it is extremely important in defining the expectations of both the professor and the student. Therefore, in order to create the most favorable learning environment, the professor reserves the right to modify or supplement the contents of this syllabus and will notify all students if it becomes necessary. It is the student's responsibility, even if physically or cognitively absent, to become aware of any changes and adjust expectations accordingly.